AdaptiveMaxPool3d#

https://pytorch.org/docs/stable/generated/torch.nn.AdaptiveMaxPool3d.html#to

Description:

Applies a 3D adaptive max pooling over an input signal composed of several input planes. The output is of size Dout×Hout×WoutD_{out} \times H_{out} \times W_{out}Dout×Hout×Wout, for any input size. The number of output features is equal to the number of input planes. output_size (Union[int, None, tuple[Optional[int], Optional[int], Optional[int]]]) – the target output size of the image of the form Dout×Hout×WoutD_{out} \times H_{out} \times W_{out}Dout×Hout×Wout. Can be a tuple (Dout,Hout,Wout)(D_{out}, H_{out}, W_{out})(Dout,Hout ,Wout) or a single DoutD_{out}Dout for a cube Dout×Dout×DoutD_{out} \times D_{out} \times D_{out}Dout×Dout×Dout. DoutD_{out}Dout, HoutH_{out}Hout and WoutW_{out}Wout can be either a int, or None which means the size will be the same as that of the input. return_indices (bool) – if True, will return the indices along with the outputs. Useful to pass to nn.MaxUnpool3d. Default: False Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Function Signature

```
class torch.nn.AdaptiveMaxPool3d(output_size,
return_indices=False)[source]#

Parameters

Shape:

forward(input)[source]#
```

Code Examples

```
>>> # target output size of 5x7x9
>>> m = nn.AdaptiveMaxPool3d((5, 7, 9))
>>> input = torch.randn(1, 64, 8, 9, 10)
>>> output = m(input)
>>> # target output size of 7x7x7 (cube)
>>> m = nn.AdaptiveMaxPool3d(7)
>>> input = torch.randn(1, 64, 10, 9, 8)
>>> output = m(input)
>>> # target output size of 7x9x8
>>> m = nn.AdaptiveMaxPool3d((7, None, None))
>>> input = torch.randn(1, 64, 10, 9, 8)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies a 3D adaptive max pooling over an input signal composed of several input planes.

The output is of size Dout×Hout×WoutD_{out} \times H_{out} \times W_{out}Dout×Hout×Wout, for any input size. The number of output features is equal to the number of input planes.

output_size (Union[int, None, tuple[Optional[int], Optional[int], Optional[int]]]) – the target output size of the image of the form Dout×Hout×WoutD_{out} \times H_{out} \times W_{out}Dout×Hout×Wout. Can be a tuple (Dout,Hout,Wout)(D_{out}, H_{out}, W_{out})(Dout,Hout,Wout) or a single DoutD_{out}Dout for a cube Dout×Dout×DoutD_{out} \times D_{out} \times D_{out}Dout×Dout×Dout. DoutD_{out}Dout, HoutH_{out}Hout and WoutW_{out}Wout can be either a int, or None which means the size will be the same as that of the input.

return_indices (bool) – if True, will return the indices along with the outputs. Useful to pass to nn.MaxUnpool3d. Default: False

Input: (N,C,Din,Hin,Win)(N, C, D_{in}, H_{in}, W_{in})(N,C,Din,Hin,Win) or (C,Din,Hin,Win)(C, D_{in}, H_{in}, W_{in})(C,Din,Hin,Win).

Output: (N,C,Dout,Hout,Wout)(N, C, D_{out}, H_{out}, W_{out})(N,C,Dout,Hout,Wout) or (C,Dout,Hout,Wout)(C, D_{out}, H_{out}, W_{out})(C,Dout,Hout,Wout), where (Dout,Hout,Wout)=output_size(D_{out}, H_{out}, W_{out})=\text{output_size}(Dout,Hout,Wout)=output_size.

Runs the forward pass.
